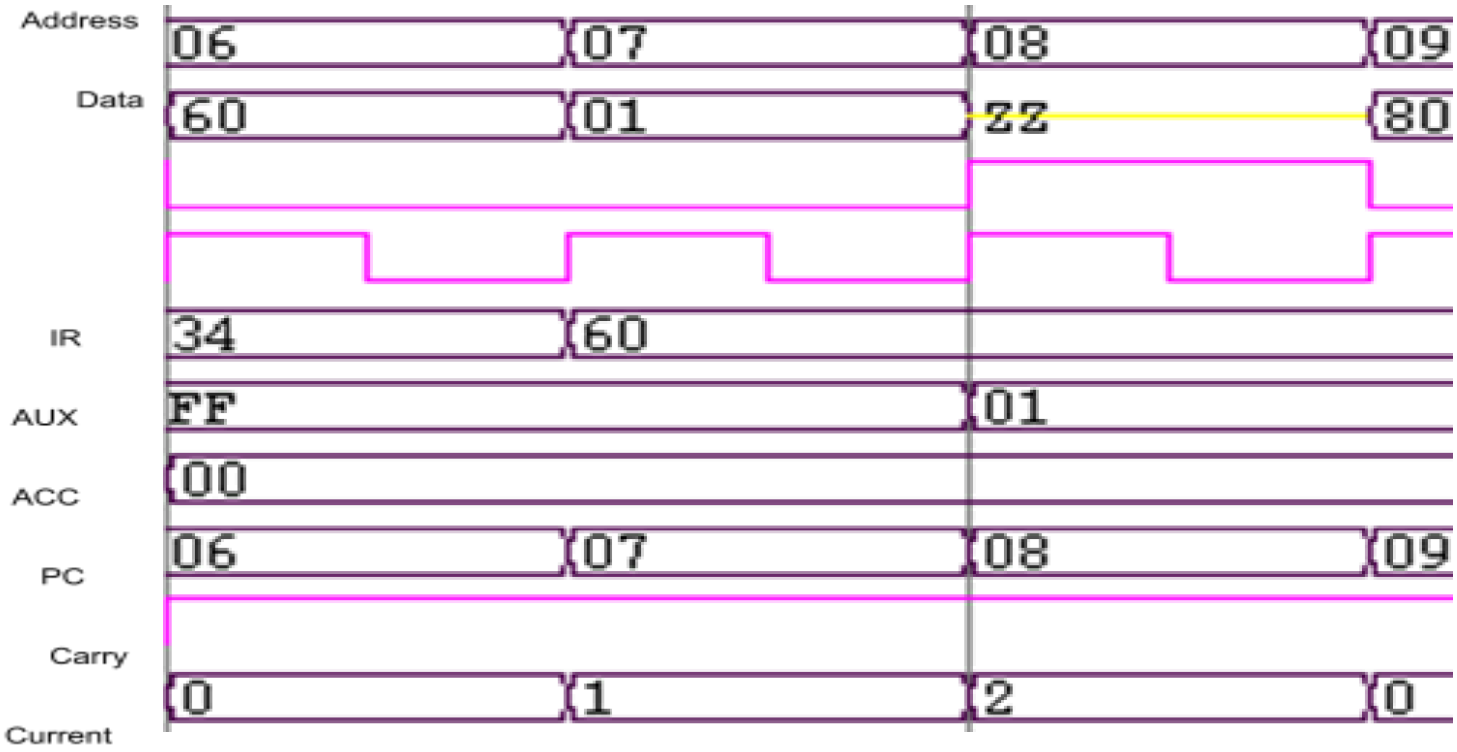
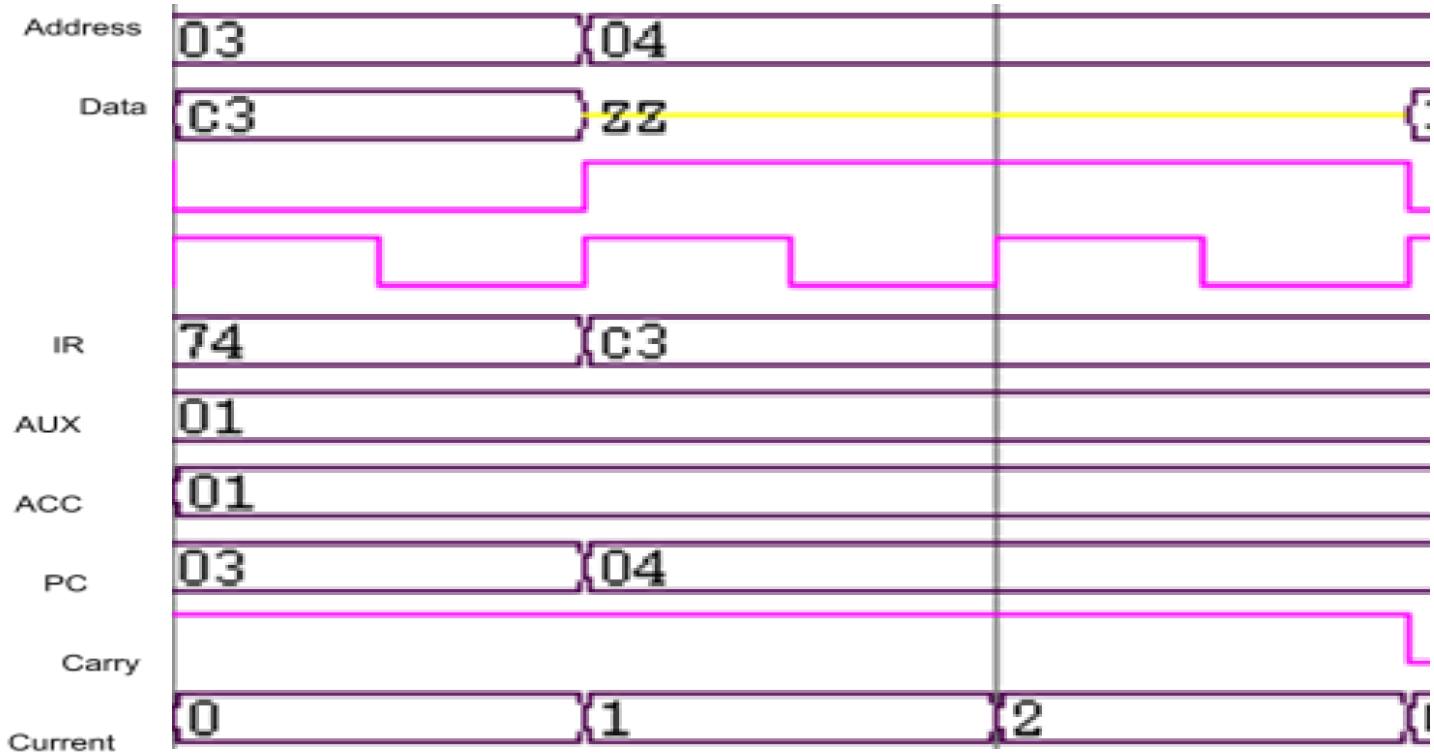
Given the following timing signals for the WIMP51 for an instruction cycle. For the Current State signal, note that 0 = fetch state, 1 = decode state, 2 = execute state.



Determine the instruction being executed during the instruction cycle.

What is the PC after executing this instruction?

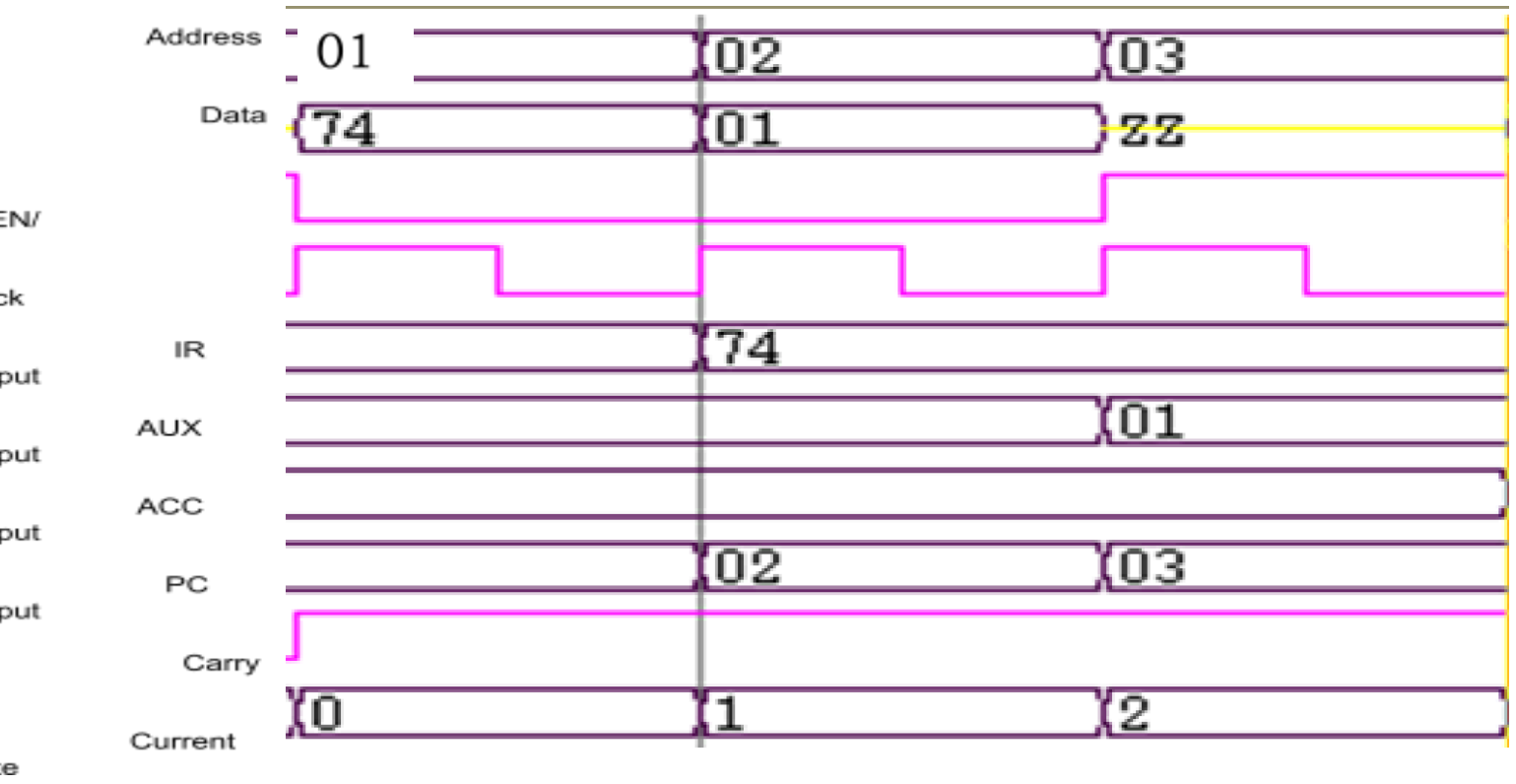
Given the following timing signals for the WIMP51 for an instruction cycle. For the Current State signal, note that 0 = fetch state, 1 = decode state, 2 = execute state.



Determine the instruction being executed during the instruction cycle.

Why does the data bus contain the value zz during the execute portion of the instruction cycle?

Given the following timing signals for the WIMP51 for an instruction cycle. For the Current State signal, note that 0 = fetch state, 1 = decode state, 2 = execute state.



Determine the instruction being executed during the instruction cycle.

Summary of WIMP51 Inst Set

ASM code	Machine code	Meaning
MOV A, #D	01110100 dddddddd	A<=D
ADDC A,#D	00110100 dddddddd	C,A<=A+D+C
MOV Rn, A	11111nnn	Rn<=A
MOV A, Rn	11101nnn	A<=Rn
ADDC A, Rn	00111nnn	C,A<=A+Rn+C
ORL A, Rn	01001nnn	A<=A OR Rn
ANL A, Rn	01011nnn	A<= A AND Rn
XRL A, Rn	01101nnn	A<=A XOR Rn
SWAP A	11000100	$A<=A_{(3-0)}\&A_{(7-4)}$
CLR C	11000011	C<=0
SETB C	11010011	C<=1
SJMP rel	10000000 aaaaaaaa	PC<=PC+rel+2
JZ rel	01100000 aaaaaaaa	PC<=PC+rel+2 if Z=1

